THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 2

Chapter 4: **Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The discriminant of $3x^2 + 2x + 5 = 0$ is: 1

(a) -55 **(b)** -54 **(c)** -57 **(d)** -56

2. The nature of the roots of $3x^2 + 2x + 1 = 0$ is: 1

(a) two distinct real roots **(b)** one real root and one imaginary root

(c) two equal real roots **(d)** no real roots

3. For the equation $x^2 - 11x + 28 = 0$, the sum of the roots is: 1

(a) 11 **(b)** 13 **(c)** 12 **(d)** 10

4. For $x^2 - 10x + k = 0$ to have equal roots, $k =$ 1

(a) 24 **(b)** 27 **(c)** 25 **(d)** 26

5. If one root of $x^2 - 13x + k = 0$ is 4, then k is: 1

(a) 36 **(b)** 37 **(c)** 35 **(d)** 38

6. The roots of $x^2 - 11x + 18 = 0$ are: 1

(a) -2 and -9 **(b)** 2 and 9 **(c)** 2 and -9 **(d)** -2 and 9

7. The discriminant of $x^2 - 9x + 20 = 0$ is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 0

8. The nature of the roots of $x^2 - 10x + 25 = 0$ is: 1

(a) one real root and one imaginary root **(b)** no real roots

(c) two equal real roots **(d)** two distinct real roots

9. For the equation $x^2 - 8x + 15 = 0$, the sum of the roots is: 1

(a) 8 **(b)** 7 **(c)** 9 **(d)** 10

10. For $x^2 - 12x + k = 0$ to have equal roots, $k =$ 1

(a) 35 **(b)** 37 **(c)** 36 **(d)** 38

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the roots of $2x^2 - 5x - 3 = 0$ by factorisation. 2

12. Find k for which $x^2 - 10x + k = 0$ has two equal roots. 2

13. Represent the situation as a quadratic equation: The product of two consecutive positive integers is 110. 2

14. Find the roots of $3x^2 + x - 2 = 0$ by factorisation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the roots of $3x^2 - 5x + 1 = 0$ using the quadratic formula (if they exist). 3

16. Solve by completing the square: $2x^2 - 7x + 3 = 0$. 3

17. Find the values of k for which $x^2 + kx + 16 = 0$ has equal roots. 3

OR

Find the roots of $2x^2 + 5x - 3 = 0$ using the quadratic formula (if they exist). 3

18. Solve by completing the square: $x^2 - 4x - 12 = 0$. 3

OR

Find the values of k for which $x^2 + kx + 25 = 0$ has equal roots. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A train travels 240 km at a uniform speed. If the speed had been 4 km/h more, it would have taken 2 hours less for the journey. Find the original speed. 5

OR

The length of a rectangular plot exceeds its breadth by 5 m and its area is 500 m². Find its dimensions. 5

20. The altitude of a right triangle is 7 cm less than its base and the hypotenuse is 17 cm. Find the base and altitude. 5

OR

The sum of the ages of two friends is 42 years. 5 years ago the product of their ages was 175. Find their present ages. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND